

RESEARCH ARTICLE

Tetraphenylsilane-Engineered MR-TADF Emitters for Pure-Green OLEDs with High-Efficiency, Long Lifetime, and Gamut Close to BT.2020 Standard

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ABSTRACT

Achieving the concurrent combination of superior color purity, high efficiency, and long operational durability remains a formidable challenge for organic light-emitting diodes (OLEDs), which are indispensable for realizing ultra-high-definition (UHD) displays. Herein, we propose a rational molecular design strategy that peripherally functionalizes the multi-resonance thermally activated delayed fluorescence (MR-TADF) skeleton with tetraphenylsilane (TPS) and deuterated TPS groups to address these challenges. The resulting emitters, *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d, exhibit sharp green emission (515–517 nm) with ultra-narrow full-widths at half-maximum (FWHMs) below 0.070 eV, ranking among the narrowest green MR-TADF materials reported. OLEDs employing *m*-DBFSi and *m*-DBFSi-d as emitters deliver the purest green electroluminescence ($\text{CIE}_y = 0.76$), extraordinary maximum external quantum efficiencies (EQE_{max}) of 40.1% and 41.9%, and maximum power efficiency (PE_{max}) approaching 200 lm W^{-1} , without employing an additional sensitizer, positioning them among the state-of-the-art narrowband pure-green OLEDs. Furthermore, the *m*-DBFSi-d-based device exhibits remarkably enhanced operational stability with operational lifetime (LT_{80}) of 2491 h at initial luminance of 1000 cd m^{-2} . Collectively, this work establishes a TPS- and deuteration-engineered MR-TADF strategy that enables OLEDs to combine ultra-narrowband emission, outstanding efficiency, and exceptional durability, paving the way for next-generation UHD displays.

1 | Introduction

Organic light-emitting diodes (OLEDs) technology has been widely adopted in high-end smartphones, televisions, and has shown great potential for near-eye augmented-reality/virtual-reality (AR/VR) displays owing to its self-emissive pixels, high contrast, and form-factor flexibility [1–4]. To support next-generation ultra-high-definition (UHD) displays, the International Telecommunication Union Radiocommunication Sector (ITU-R) announces the Broadcasting and Television 2020 (BT.2020) color-gamut standard, which defines the Commission

Internationale de l'Éclairage (CIE) coordinates for the three primary colors [5, 6]. Among them, the green vertex at (0.170, 0.797) represents the most stringent and technologically demanding target because it requires emission precisely centered in the green region with an exceptionally narrow full-width at half-maximum (FWHM) [7, 8]. Multi-resonance thermally activated delayed fluorescence (MR-TADF) materials based on boron-nitrogen (BN) frameworks have been widely studied as promising emitters, owing to the intrinsically narrow FWHM and theoretically unity exciton harvesting [9–11]. However, for the green subpixel, which strongly affects both color-gamut coverage and perceived

brightness, simultaneously achieving BT.2020-level color purity, high external quantum efficiency, and long operational lifetime within a single emitter system remains highly challenging, thereby motivating the development of new organic green emitters [12–21]. Among the known blue MR-TADF archetypes, *v*-DABNA reported by Hatakeyama and co-workers stands as a landmark compound with ultra-narrow FWHM of 14 nm and rigorously localized MR framework [22]. Guided by this paradigm, subsequent efforts have expanded the molecular diversity by elongating the π -conjugated skeleton to redshift the emission into the green region while preserving the MR characteristic, and/or by appending peripheral substituents that further refine the emission curve, mitigate aggregation, suppress vibronic coupling, and enhance the excited-state stability [23–30]. Despite extensive molecular engineering, most green MR-TADF derivatives exhibit CIE_y below 0.75. In an effort to overcome this limitation, Ge et al. [31] reported a series of “bifunctional-group” modulated emitters (DBNDS-TPh, DBNDS-DFPh, and DNNDs-CNPh) that impressively realized high-color-purity green emissions (CIE_y = 0.75/0.76) and external quantum efficiency (EQE) of up to 35%. However, the operational stability of their devices remains a critical bottleneck. Even with phosphor sensitization in a ternary EML (emitting layer) configuration, the LT₅₀s (defined as the time for luminance to decay to 50% of its initial value) of their devices at an initial luminance of 1000 cd m^{−2} were less than 200 h, far from the requirement for practical applications. Despite considerable progress, the realization of pure-green MR-TADF emitters that concurrently deliver narrowband emission, high efficiency, and long-term operational stability remains a critical yet unresolved challenge.

Tetraphenylsilane (TPS) has been widely recognized as a robust building block for OLED host materials due to its excellent morphological stability, high thermal resistance, and high triplet energy [32–38]. The sp³-hybridized Si–C bonds enforce a typical tetrahedral configuration that effectively suppresses intermolecular π – π stacking, mitigates aggregation-caused quenching (ACQ), and reduces exciton–exciton and exciton–polaron annihilation, thereby collectively suppressing nonradiative decay channels and enabling efficient and stable electroluminescence (EL) [39–43]. Previous studies have demonstrated that introducing TPS units into host frameworks significantly enhances device lifetime and efficiency by attenuating excited-state intermolecular coupling and stabilizing the emissive microenvironment. For example, Kim et al. [36] demonstrated that a TPS-based exciplex host prolonged the LT₉₅ of blue phosphorescent OLEDs based on platinum complex (BD-02) at an initial luminance of 1000 cd m^{−2} by approximately 50 times, while Park et al. [37] developed a TPS-contained host (mTDBA-2Si) that enabled a three-fold enhancement in EL efficiency (EQE_{max} = 38.1%). These results unambiguously demonstrate the multifunctionality of TPS units in suppressing excited-state interactions and enhancing both efficiency and durability of electroluminescent devices. Despite these advantages, direct integration of TPS into the emissive core of MR-TADF emitters has only been tentatively explored [44, 45]. Given its steric bulk and good solubility, herein, we develop a molecular design strategy that synergistically combines π -extension and TPS substituent to construct pure-green MR-TADF emitters delivering superior color purity, high efficiency, and prolonged operational stability (Figure 1). Specifically, the dibenzofuran (DBF) unit is embedded into a *meta*-positioned

diboron framework to narrow the bandgap (E_g) and shift the emission into the green region, while tetrahedral TPS groups are linked to the *para*-positions of boron atoms to provide 3D protection against detrimental interaction. Two positional isomers, *p*-DBFSi and *m*-DBFSi, are obtained, and subsequent deuteration of TPS groups yields *m*-DBFSi-d, in which the strengthened C–D bonds and reduced zero-point energy (ZPE) effectively suppress degradation pathways and molecular vibration [46–50]. These emitters exhibit pure-green emission (515–517 nm) with ultra-small FWHM values below 15 nm (0.070 eV), ranking among the sharpest green MR-TADF emitters reported. Combining emission peaks around 520 nm and suppressed vibronic shoulder, the devices employing *m*-DBFSi and *m*-DBFSi-d achieve superior EL performance, with a CIE_y value of 0.76 (approaching the BT.2020 green standard), along with exceptional EQE_{max} of 40.1% and 41.9%, respectively, and maximum power efficiency (PE_{max}) close to 200 lm W^{−1} (187.5 and 191.8 lm W^{−1}, respectively). Moreover, the deuterated analogue *m*-DBFSi-d exhibits a prolonged operational lifetime (T₈₀) of 2491 h at an initial luminance of 1000 cd m^{−2}, outperforming its non-deuterated counterpart *m*-DBFSi (2006 h) and all reported green emitters based on the *meta*-diboron system. These results demonstrate that incorporating TPS units, especially in their deuterated form, effectively stabilizes the excited state and mitigates degradation processes, offering a viable route to achieve simultaneously high color purity, efficiency, and operational stability in MR-TADF emitters for ultra-high-definition green OLEDs.

2 | Results and Discussion

2.1 | Molecular Design, Synthesis, and Characterization

The molecular design and chemical structures of *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d are depicted in Figure 1. To realize pure-green emission, a π -conjugation extension strategy is employed by embedding dibenzofuran (DBF) units into the *meta*-diboron framework, which effectively narrows E_g and shifts the emission toward the green spectral region. To further tailor the emission curve and enhance molecular robustness, TPS substituents are introduced at the *para*-positions of the boron atoms, yielding *p*-DBFSi. In parallel, the isomer *m*-DBFSi, in which triphenylsilyl groups are linked at *meta*-positions of the MR skeleton, is designed to refine the steric topology for TPS incorporation. This structural modification strengthens the 3D shielding conferred by the tetrahedral TPS units, thereby suppressing molecular packing and detrimental intermolecular interactions. To further enhance the molecular stability, a partially deuterated analog *m*-DBFSi-d is simply synthesized by introducing deuterated TPS groups. The reduced zero-point energy (ZPE) and increased C–D bond dissociation energy (BDE) relative to the C–H analogue is expected to suppress vibration, attenuate non-radiative decay and reinforce the molecular backbone. Together, the combined effects of π -extension, steric-topology modulation, and isotopic substitution are intended to realize ultra-pure green MR-TADF emitters with high efficiency and improved operational stability.

The synthesis routes of target compounds are demonstrated in Schemes S1 and S2. The key intermediate DBFCl was

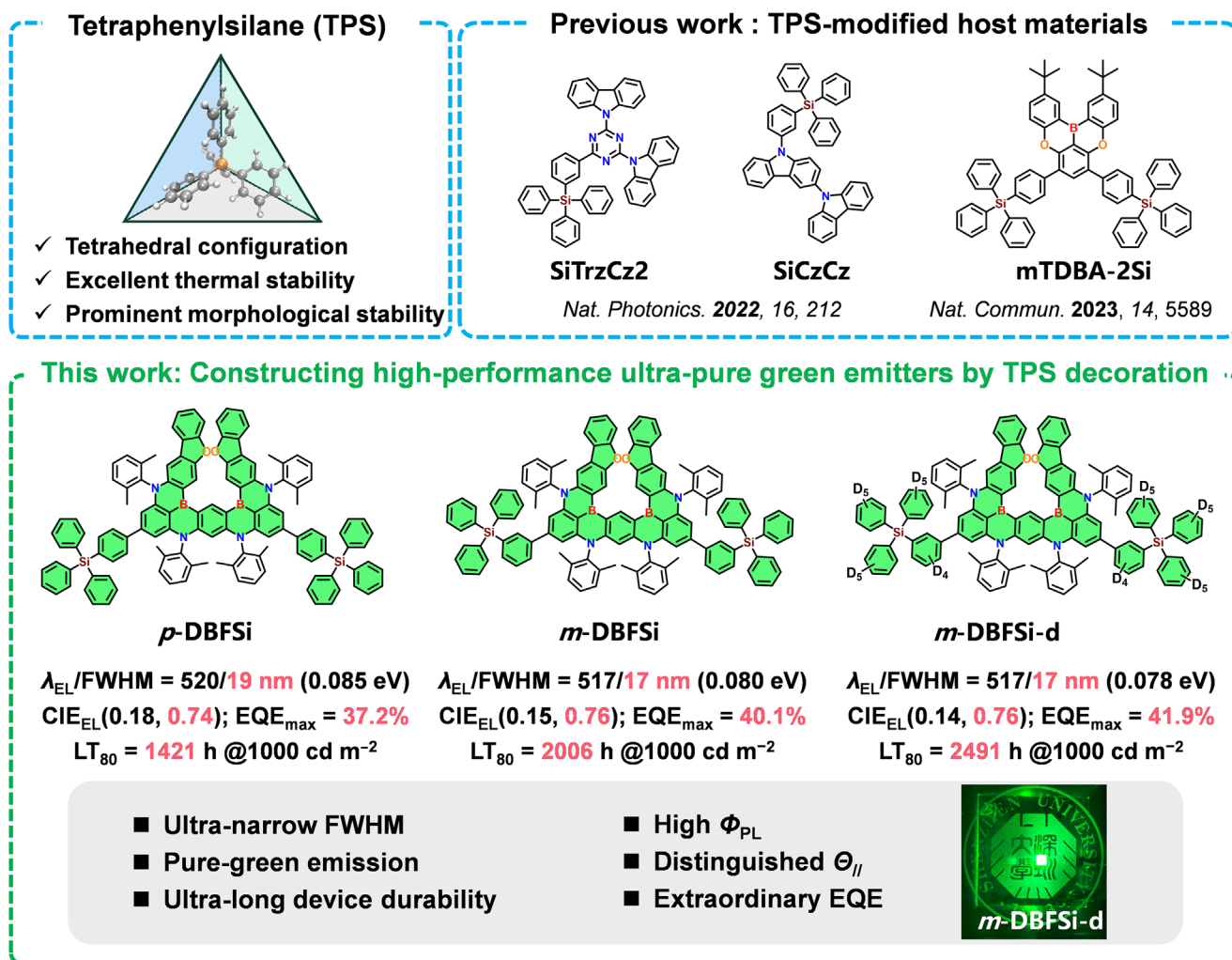


FIGURE 1 | Chemical structure and molecular design strategy of pure-green MR-TADF emitters.

synthesized in high yield through a one-shot regioselective borylation reaction [51]. The target molecules *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d were subsequently synthesized via Suzuki-Miyaura coupling of DBFCl with commercially available (4-(triphenylsilyl)phenyl)boronic acid, triphenyl(3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl)silane, and tris(phenyl-*d*₅)(3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl-1,2,4,5,6-*d*₄)silane, respectively. All target compounds were purified through column chromatography followed by recrystallization, and their chemical structures were verified by ¹H, ¹³C nuclear magnetic resonance (NMR) spectroscopy as well as high-resolution mass spectrometry (HRMS) (Figures S1–S19). Single crystals (Figure S20 and Tables S1 and S2) of *p*-DBFSi and *m*-DBFSi were successfully obtained to further confirm their structures by slowly diffusing ethanol into their respective dichloromethane solutions. The crystallographic analyses reveal that the MR skeleton adopts a quasi-planar configuration, favorable for horizontal dipole orientation and efficient light out-coupling in OLEDs. The nearly perpendicular 2,6-dimethylphenyl substituents and the tetrahedral TPS units provide steric hindrance to help suppress intermolecular π – π stacking. Compared with the *para*-linked derivative, the *meta*-linked TPS units in *m*-DBFSi exhibit a larger deviation from the mean plane of the MR skeleton, leading to a more steric

molecular geometry and looser intermolecular packing, fully consistent with our molecular design.

2.2 | Simulated Calculations

To investigate the structural features and electronic properties of these emitters, density functional theory (DFT) and time-dependent DFT (TD-DFT) calculations were performed at the B3LYP/6-31G (d, p) level. As depicted in Figure 2 and Figure S21, the frontier molecular orbitals (FMOs) of all emitters exhibit an alternating distribution pattern at the atomic level, characteristic of the multi-resonance (MR) effect that underpins their narrowband emission. Upon peripheral functionalization with TPS moieties, the lowest unoccupied molecular orbital (LUMO) is delocalized onto the TPS phenyl rings directly attached to the MR core, which narrows the E_g and induces a red-shifted emission spectrum. Quantitative orbital partitioning reveals that the FMOs remain essentially confined to the MR core (HOMO: ca. 99.6%; LUMO: ca. 89%–91%), while the distal Ph₃Si substituent contributes only marginally to the LUMO (at most 1.2%; Table S3). This confinement is indicative of σ -interrupted coupling at the tetrahedral *sp*³-Si junction, which suppresses through-bond delocalization beyond the bridge phenyl. Consistently,

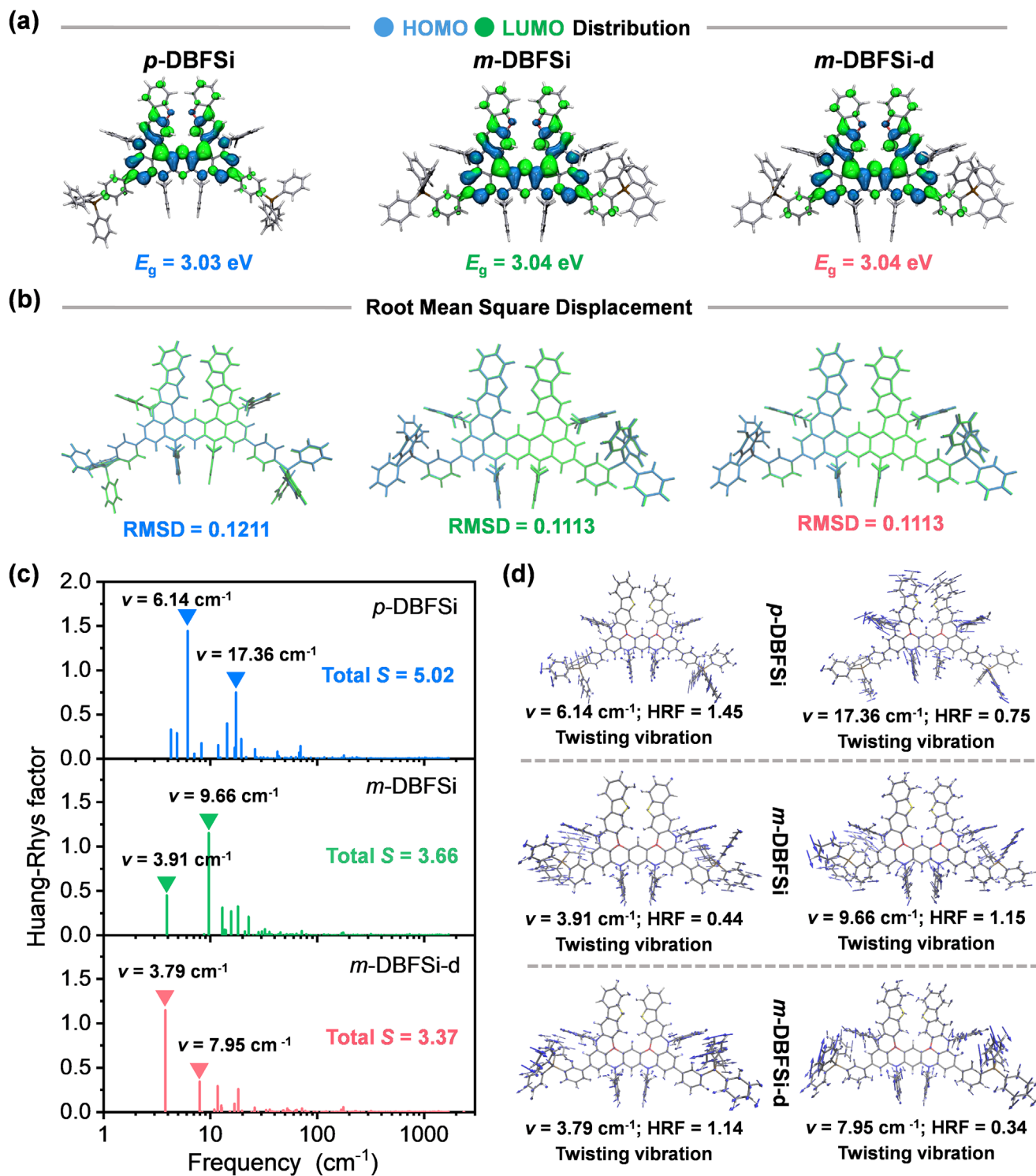


FIGURE 2 | (a) The frontier molecular orbital distributions of *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d. (b) The root-mean-square-displacement (RMSD) values of *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d. (c) The Huang–Rhys factors of *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d under different vibrational modes. (d) The representative vibration modes for S_1-S_0 transition of *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d.

fragment-resolved hole-electron overlap analysis and transition density matrix (TDM) analysis indicate that the S_1 excitation remains predominantly core-centered with pronounced overlap and negligible involvement of the distal Ph_3Si unit (Figure S22) [52–55]. As a structural consequence, the 3D topology of TPS groups serves a dual role: it suppresses π – π stacking interactions and enhances morphological and thermal stability. Moreover,

by acting as a bulky and chemically inert substituent, TPS could mitigate Dexter energy transfer processes, thereby reducing exciton loss and improving device operational durability. Subtle orbital distinctions are observed between the positional isomers. In *p*-DBFSi, the C–Si bond lies within the LUMO distribution region, and owing to the weak electron-withdrawing character of the silicon atom, the LUMO energy level (–1.65 eV) is slightly

lower than those of *m*-DBFSi and *m*-DBFSi-d (−1.60 eV). This small yet non-negligible perturbation is expected to manifest as a further redshift of the emission peak. The calculated HOMO and LUMO energy levels are −4.68/−1.64 eV for *p*-DBFSi, −4.64/−1.60 eV for *m*-DBFSi, and −4.64/−1.60 eV for *m*-DBFSi-d, respectively. Moreover, the relatively large oscillator strengths ($f > 0.4$) obtained for all emitters point to highly efficient radiative transition channels, which facilitate efficient exciton utilization and high photoluminescence quantum yields (Φ_{PL} s).

To probe how geometry shapes the spectra, we quantified structural relaxation using the root-mean-square displacement (RMSD) between S_0 and S_1 and the reorganization energies (λ_{S_0} and λ_{S_1}), evaluated with Nelson's four-point method (Figure 2b; Figure S23) [56]. All emitters display small RMSD and reorganization energy values, consistent with a rigid MR framework, namely 0.1211 Å/0.1061 eV for *p*-DBFSi, 0.1113 Å/0.0952 eV for *m*-DBFSi, and 0.1113 Å/0.0935 eV for *m*-DBFSi-d. These values indicate that the excited-state potential energy surface closely follows the ground-state surface, which minimizes geometric distortion upon excitation and favors narrow emission bandwidths. Notably, the *meta*-linked TPS architecture in *m*-DBFSi and *m*-DBFSi-d produces slightly smaller RMSD and λ_{total} values than the *para*-analogue, stabilizing a more rigid excited-state landscape that is typically associated with narrower emission bands and better color purity. To further elucidate how vibrational dynamics shape the spectra, we evaluated the Huang–Rhys factors (S) and $\lambda_{S_0}/\lambda_{S_1}$ contributions for individual vibrational modes using the Molecular Materials Property Prediction Package (MOMAP) [57, 58]. As shown in Figure 2c and Figure S24, all molecules exhibit largely similar vibrational fingerprints that are dominated by low-frequency torsion of the backbone, while high-frequency stretching modes contribute weakly. For *p*-DBFSi, the principal modes occur at 6.14 cm^{-1} ($S = 1.45$) and 17.36 cm^{-1} ($S = 0.75$); for *m*-DBFSi, at 3.91 cm^{-1} ($S = 0.44$) and 17.36 cm^{-1} ($S = 1.15$); and for *m*-DBFSi-d, at 3.79 cm^{-1} ($S = 1.14$) and 7.95 cm^{-1} ($S = 0.34$). When summed over all modes above, *m*-DBFSi exhibits a smaller total Huang–Rhys factor ($S_{\text{total}} = 3.66$) than *p*-DBFSi ($S_{\text{total}} = 5.02$), and partial deuteration further lowers S_{total} to 3.37 in *m*-DBFSi-d. In parallel, fragment-resolved normal-mode projection demonstrates a pronounced reduction in the HRF-weighted MR-core contribution within the dominant vibronic modes upon TPS attachment (Figure S25), suggesting that the dominant vibronic activity is progressively redistributed toward low-frequency torsional motions of the peripheral substituents. Vibrational-vector analyses (Figure 2d) provide additional insight into the nature of these modes. Taken together, the reduced S_{total} and reorganization energies are diagnostic of weaker vibronic coupling between S_1 and S_0 , which is consistent with a narrower FWHM and improved color purity. Shifting vibronic activity to these weakly electronically coupled peripheries channels the relaxation into softer and lower-frequency modes, thereby weakening electron-phonon (vibronic) coupling at the radiative transition and damping nonradiative channels. Partial deuteration in *m*-DBFSi-d further amplifies this effect by replacing high-frequency C–H modes with lower-frequency C–D vibrations. The associated reduction in zero-point energy and downshift of vibrational frequency make multi-phonon pathways for bond cleavage and hydrogen abstraction less accessible, which rationalizes the expectation of slower chemical degradation and improved operational lifetime [46–50]. Taken together, the RMSD/ λ_{total} metrics, the mode-

resolved Huang–Rhys analysis, and the vibrational-vector maps consistently indicate that *meta*-linked TPS engineering, further amplified by partial deuteration, establishes a rigid yet electronically faithful framework that limits structural relaxation, weakens vibronic coupling, and provides a solid basis for narrowband green emission with high Φ_{PL} , color-purity, and device durability.

2.3 | Photophysical Properties

The photophysical properties of these emitters were investigated in dilute toluene solutions at a concentration of 1.0×10^{-5} M at room temperature (Figure 3a and Table 1). Intense and distinct absorption peaks are observed at 506, 504, and 504 nm for *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d, respectively, which are assigned to short-range charge transfer (SRCT) transitions, a feature of typical MR-type molecules. The photoluminescence (PL) spectra exhibit sharp green emission peaking at 517, 515, and 515 nm for *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d, respectively, each exhibiting exceptionally narrow FWHM values of less than 0.070 eV. Specifically, the corresponding FWHM values are 15 nm (0.067 eV), 14 nm (0.065 eV), and 14 nm (0.064 eV) for *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d, respectively, representing the smallest FWHM values reported for green MR-TADF emitters (Table S4). The minimal Stokes shifts (11 nm) further corroborate the suppressed structural relaxation between the S_1 and S_0 states, aligning well with theoretical predictions. Additionally, based on the low-temperature fluorescence and phosphorescence spectra recorded at 77 K, the singlet-triplet energy gap (ΔE_{ST}) values of *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d are determined to be 0.10 eV (Figure S26), such small ΔE_{ST} values facilitate efficient exciton up-conversion from the T_1 to S_1 states, a critical requirement for high-performance TADF emitters. To gain deeper insight into the photophysical properties of the compounds in the solid state, vacuum-deposited films doped with 3 wt% of these emitters were fabricated using 5-(3-(4,6-diphenyl-1,3,5-triazin-2-yl)phenyl)-7,7-dimethyl-5,7-dihydroindeno[2,1-b]carbazole (DMIC-TRZ) as the host [59]. The Φ_{PL} values of these emitters in the corresponding vacuum-deposited films are determined to be 96%, 99%, and 99% for *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d, respectively. The nearly unity Φ_{PL} values of the *m*-DBFSi- and *m*-DBFSi-d-doped films are attributed to the rigid molecular frameworks and suppressed structural relaxation, which effectively reduce nonradiative decay channels. The emission peaks of *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d are observed at 520, 518, and 518 nm, respectively, exhibiting a subtle red shift compared to their dilute toluene solutions, which could be attributed to the increased polarity of the surrounding matrix and intermolecular interactions. Benefiting from the stronger aggregation-suppressing characteristics, the *m*-DBFSi and *m*-DBFSi-d films show only minimal spectral broadening, with FWHMs of 17 nm (0.078 eV) and 17 nm (0.076 eV), respectively. Notably, the *m*-DBFSi and *m*-DBFSi-d doped films achieve CIE coordinates of (0.15, 0.76) and (0.14, 0.76), which are very close to the BT.2020 green standard (0.17, 0.80). In contrast, stronger intermolecular interactions in *p*-DBFSi lead to slight spectral broadening, accompanied by a slightly increased shoulder feature, resulting in a FWHM of 19 nm (0.086 eV). Although its emission peak is marginally closer to the pure-green region, the corresponding coordinates of (0.20, 0.74) inferior to *m*-DBFSi and *m*-DBFSi-d underscore the critical

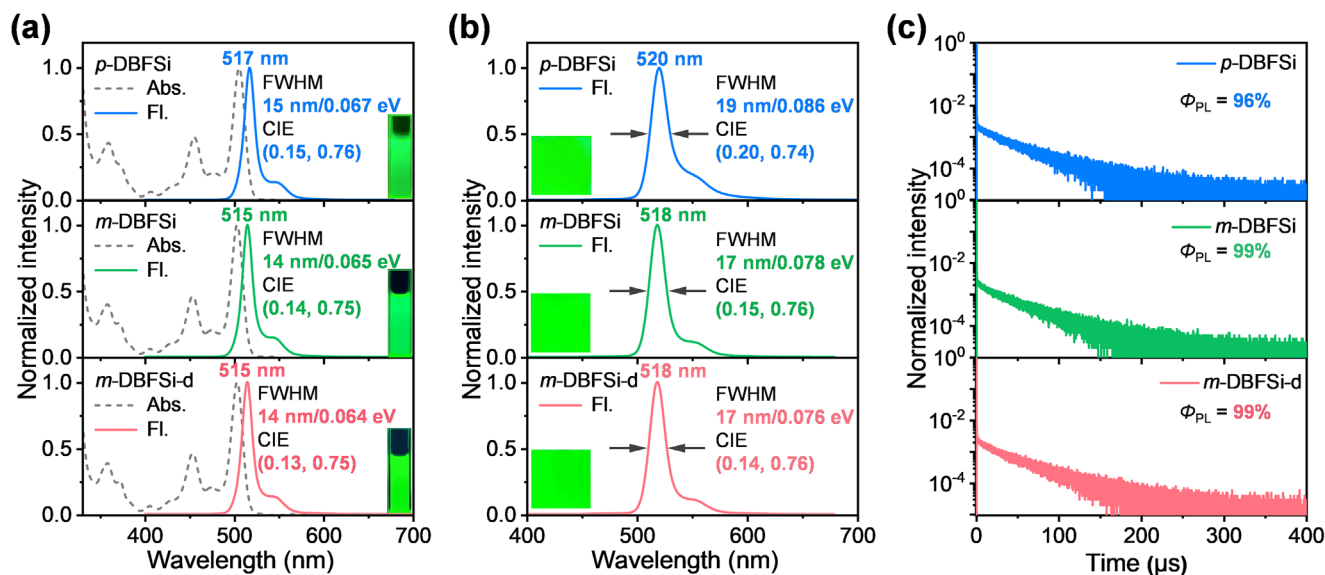


FIGURE 3 | Photophysical properties of emitters. (a) The UV-vis spectra and fluorescence spectra of *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d in diluted toluene (1.0 × 10⁻⁵ M) (Insets show the corresponding solution photographs under 365 nm UV light irradiation). (b) The fluorescence spectra of *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d in 3 wt% DMIC-TRZ doped films (Insets show the corresponding film photographs under 365 nm UV light irradiation). (c) Transient photoluminescence decay curves in 3 wt% DMIC-TRZ doped films.

TABLE 1 | Photophysical properties of *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d.

Emitter	λ_{abs}^a [nm]	λ_{em}^a [nm]	FWHM ^a [nm/eV]	ΔE_{ST}^b [eV]	Φ_{PL}^c [%]	τ_{PF}^c [ns]	τ_{DF}^c [μs]	$k_{\text{r,S}}^c$ [10 ⁷ s ⁻¹]	$k_{\text{nr,S}}^c$ [10 ⁵ s ⁻¹]	k_{RISC}^c [10 ⁵ s ⁻¹]	$\Theta_{//}^d$ [%]
<i>p</i> -DBFSi	506	517	15/0.067	0.10	96	6.7	42.1	2.2	8.9	1.6	98
<i>m</i> -DBFSi	504	515	14/0.065	0.10	99	6.3	40.5	3.6	3.8	1.1	97
<i>m</i> -DBFSi-d	504	515	14/0.064	0.10	99	5.8	39.4	3.8	3.7	1.2	97

^a) Measured in dilute toluene (1 × 10⁻⁵ M);

^b) $\Delta E_{\text{ST}} = E_{\text{S}_1} - E_{\text{T}_1}$, the singlet and triplet energy levels were determined from the onsets of fluorescence and phosphorescence spectra recorded at 77 K in toluene solution;

^c) Measured in 3 wt% doped DMIC-TRZ film;

^d) Determined from angle-dependent p-polarized photoluminescence spectra in 3 wt% doped DMIC-TRZ film.

importance of attaining an ultra-narrow FWHM to ensure high color purity. To evaluate the concentration tolerance enabled by TPS, concentration-dependent PL spectra were measured using DBFBN (without TPS) as a control in DMIC-TRZ doped films (Figure S27). The DBFBN films exhibit an obvious red-shift and spectral broadening with a pronounced Φ_{PL} decrease at higher doping concentrations, whereas *m*-DBFSi maintains nearly invariant emission profiles with only minor Φ_{PL} reduction, indicating suppressed ACQ.

Transient PL measurements on the 3 wt% doped films confirm the TADF character at room temperature (Figure 3c; Figure S28). All samples exhibit clear bi-exponential decay profiles, comprising nanosecond-scale prompt fluorescence (τ_{PF}) and microsecond-scale delayed fluorescence (τ_{DF}) components. The τ_{PF} and τ_{DF} are 6.7 ns/42.1 μs, 6.3 ns/40.5 μs, and 5.8 ns/39.4 μs for *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d, respectively. The short τ_{PF} values, together with high Φ_{PL} s, indicate a rapid radiative transition of singlet excitons. To gain a quantitative understanding of the underlying kinetic processes, key rate constants including singlet radiative decay ($k_{\text{r,S}}$), nonradiative decay ($k_{\text{nr,S}}$) and reverse inter-

system crossing (k_{RISC}) were determined according to previously reported methods (Table 1; Table S6). The corresponding $k_{\text{r,S}}$ values for *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d were calculated to be 2.2 × 10⁷ s⁻¹, 3.6 × 10⁷ s⁻¹, and 3.8 × 10⁷ s⁻¹, respectively. Meanwhile, the $k_{\text{nr,S}}$ values were 8.9 × 10⁵ s⁻¹, 3.8 × 10⁵ s⁻¹, and 3.7 × 10⁵ s⁻¹, respectively. The combination of an increased $k_{\text{r,S}}$ and a reduced $k_{\text{nr,S}}$ enables *m*-DBFSi and *m*-DBFSi-d to exhibit superior emission efficiency. In addition, *p*-DBFSi exhibits a higher k_{RISC} , which is primarily attributed to its slightly enhanced spin-orbit coupling (SOC) (Figure S30). The corresponding k_{RISC} values are measured to be 1.6 × 10⁵ s⁻¹ for *p*-DBFSi, 1.1 × 10⁵ s⁻¹ for *m*-DBFSi, and 1.2 × 10⁵ s⁻¹ for *m*-DBFSi-d, respectively, indicating notable capacity for harvesting triplet excitons. The TADF characteristics of these emitters in mCBP doped films were further verified by temperature-dependent transient PL decay curves, where the delayed fluorescence component becomes more pronounced at elevated temperatures (Figure S29). To further elucidate the dipole orientations, angle-resolved PL measurements were performed on the doped films. All emitters in DMIC-TRZ exhibit nearly perfect horizontal dipole alignment ($\Theta_{//}$ values approaching 100%; Figure S31). Such pronounced $\Theta_{//}$ values are

expected to enhance the optical out-coupling efficiency (η_{out}) and thereby contributing to high EQEs of the corresponding devices [60, 61]. Prior to device fabrication, cyclic voltammetry (CV) measurements were carried out to determine the HOMO energy levels of these emitters, which are -5.13 , -5.12 , and -5.10 eV for *p*-DBFSi, *m*-DBFSi and *m*-DBFSi-d, respectively. By combining these values with their optical bandgaps, the LUMO levels are estimated to be -2.75 , -2.72 , and -2.71 eV, respectively (Figure S32 and Table S5). Thermogravimetric analysis (TGA) confirms the excellent thermal stability of these emitters, with high decomposition temperatures (T_d , corresponding to 5% mass loss) of 516°C , 505°C , and 513°C for *p*-DBFSi, *m*-DBFSi and *m*-DBFSi-d, respectively (Figure S33). In addition, High-Performance Liquid Chromatography (HPLC) analyses performed before and after vacuum thermal evaporation show nearly unchanged chromatographic profiles with no detectable new impurity peaks and a nearly identical main-component fraction (Figures S34–S36), corroborating the thermal-evaporation stability of these emitters under device-relevant conditions. Such high thermal stability is particularly beneficial for the fabrication of vacuum-deposited OLEDs.

2.4 | OLEDs Device Performances

To evaluate the EL performances of these emitters, OLEDs were fabricated with the optimized device structure of indium tin oxide (ITO)/dipyrazino[2,3-f:2',3'-h]quinoxaline-2,3,6,7,10,11-hexacarbonitrile (HATCN, 5 nm)/ N^4,N^4'',N^4''' -tetra([1,1'-biphenyl]-4-yl)-[1,1'-biphenyl]-4,4'-diamine (BPBPA, 50 nm)/9-([1,1'-biphenyl]-3-yl)-9'-([1,1'-biphenyl]-4-yl)-9H,9'H-3,3'-bicarbazole (BiCar, 15 nm)/EML, 55 nm/(1,3,5-triazine-2,4,6-triyl)tris(benzene-3,1-diyl)tris(diphenylphosphine oxide (POT2T, 20 nm)/1-(4-(10-([1,1'-biphenyl]-4-yl)anthracen-9-yl)phenyl)-2-ethyl-1H-benzimidazole (ANT-BIZ, 30 nm)/hydroxyquinolinolato-lithium (Liq, 2 nm)/aluminum (Al, 100 nm). The architecture is illustrated in Figure 4a, ITO and Al serve as the anode and cathode, respectively. Charge injection is facilitated by HATCN (hole-injection layer, HIL) and Liq (electron-injection layer, EIL). BPBPA functions as the hole-transporting layer (HTL), while BiCar and POT2T act as the electron- and hole-blocking layers (EBL and HBL), respectively, and ANT-BIZ is employed as the electron-transporting layer (ETL). DMIC-TRZ is selected as the host material for the EML, which provides well-matched energy levels with adjacent layers, ambipolar charge transport, and a narrow singlet–triplet gap that together balance carrier transport and facilitate efficient exciton utilization. The energy level alignment, chemical structures of the functional layer materials and the EL performance are illustrated in Figure 4 and Figures S37–S39, with the corresponding key parameters summarized in Table 2 and Table S11.

Devices incorporating *p*-DBFSi, *m*-DBFSi and *m*-DBFSi-d display characteristically narrow green EL (Figure 4b), featuring peak wavelengths/FWHM values of 520 nm/19 nm (0.085 eV), 517 nm/17 nm (0.080 eV), and 517 nm/17 nm (0.078 eV), respectively. In particular, the *m*-DBFSi- and *m*-DBFSi-d-based devices afford ultra-narrow emission bands that rank among the narrowest reported for bottom-emitting green OLEDs (Figure 4f; Table S7). The combination of well-confined emission maxima, suppressed shoulder features, and minimal spectral broadening

enable CIE coordinates of (0.15, 0.76) and (0.14, 0.76), which surpass the NTSC green standard and approach the BT.2020 pure-green target (0.17, 0.80) [5, 6]. By contrast, although the *p*-DBFSi-based device maintains a relatively narrow spectrum at an appropriate peak position, its stronger shoulder results in compromised color purity, as reflected by the shifted CIE coordinates of (0.20, 0.74). All devices exhibit low turn-on voltages (V_{on}) of 2.4 V and high maximum luminance (L_{max}), which are consistent with a well-aligned energy cascade across the functional layers. The spectra remain highly stable throughout the driving voltage range of 3.2–9.8 V (Figure S38), with negligible variation in normalized EL spectra, indicating an efficient exciton recombination and energy transfer process in the EMLs. Leveraging near-unity Φ_{PL} together with high $\Theta_{\text{||}}$, the *m*-DBFSi- and *m*-DBFSi-d-based devices deliver exceptional EL performances. As illustrated in Figure 4c,d, their EQE_{max} reach 40.1% and 41.9%, respectively, establishing a new benchmark among OLEDs employing MR-TADF emitters in the pure-green region (Figure 4f; Table S7). Benefiting from the combination of low operating voltage, exceptional EQE, and ultra-narrow FWHM, both devices achieve unprecedented maximum power efficiency (PE) approaching 200 lm W^{-1} , with values of $187.5/191.8 \text{ lm W}^{-1}$, positioning them among the cutting-edge narrowband pure-green OLEDs. Additionally, both OLEDs reach considerable maximum current efficiency (CE_{max}) of $143.2/146.4 \text{ cd A}^{-1}$, respectively. By comparison, the *p*-DBFSi device exhibits slightly inferior performance, with $\text{EQE}_{\text{max}}/\text{CE}_{\text{max}}/\text{PE}_{\text{max}}$ of $37.2\%/140.7 \text{ cd A}^{-1}/165.4 \text{ lm W}^{-1}$, respectively. Beyond the enhanced out-coupling efficiency imparted by the high $\Theta_{\text{||}}$ values, the superior EL performances of *m*-DBFSi and *m*-DBFSi-d are most reasonably ascribed to the concerted effects of weaker detrimental intermolecular interactions and accelerated radiative transitions. Notably, the concentration-dependent EL behaviors provide compelling evidence for suppressed intermolecular interactions (Figure S40 and Table S10). With increasing doping concentration, the DBFBN-based devices exhibit a clear red-shift and spectral broadening, together with a pronounced decline in EQE_{max} , whereas the TPS-decorated *m*-DBFSi-based devices maintain invariant EL spectral profiles with only minor decline in EQE_{max} , demonstrating that TPS incorporation effectively weakens detrimental intermolecular interactions, thereby enabling efficient electroluminescence.

With the EL spectra of the *p*-DBFSi-, *m*-DBFSi-, and *m*-DBFSi-d-based devices being highly similar in emission peak, FWHM and vibronic structure, a direct and unbiased comparison of their intrinsic operational stability could be performed. The device lifetime increases progressively from *p*-DBFSi to *m*-DBFSi and further to *m*-DBFSi-d. Under constant-current driving at an initial luminance of 1000 cd m^{-2} (Figure S41), the LT_{50} lifetimes (time to lose 50% of the initial luminance) of *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d are 299, 371, and 450 h, respectively, which surpass all reported binary devices based on green emitters with meta-diboron frameworks (Table S7). To trace the origin of this enhanced stability, electrochemical and photostability measurements were performed (Figures S42 and S43). Among the series, *m*-DBFSi-d demonstrates the most robust behavior, maintaining nearly invariant CV profiles after 100 cycles and showing only a slight attenuation in PL intensity accompanied by a negligible chromaticity shift under continuous UV irradiation for 168 h, thereby substantiating its enhanced redox and photostability relative to *p*-DBFSi and *m*-DBFSi [62, 63]. This

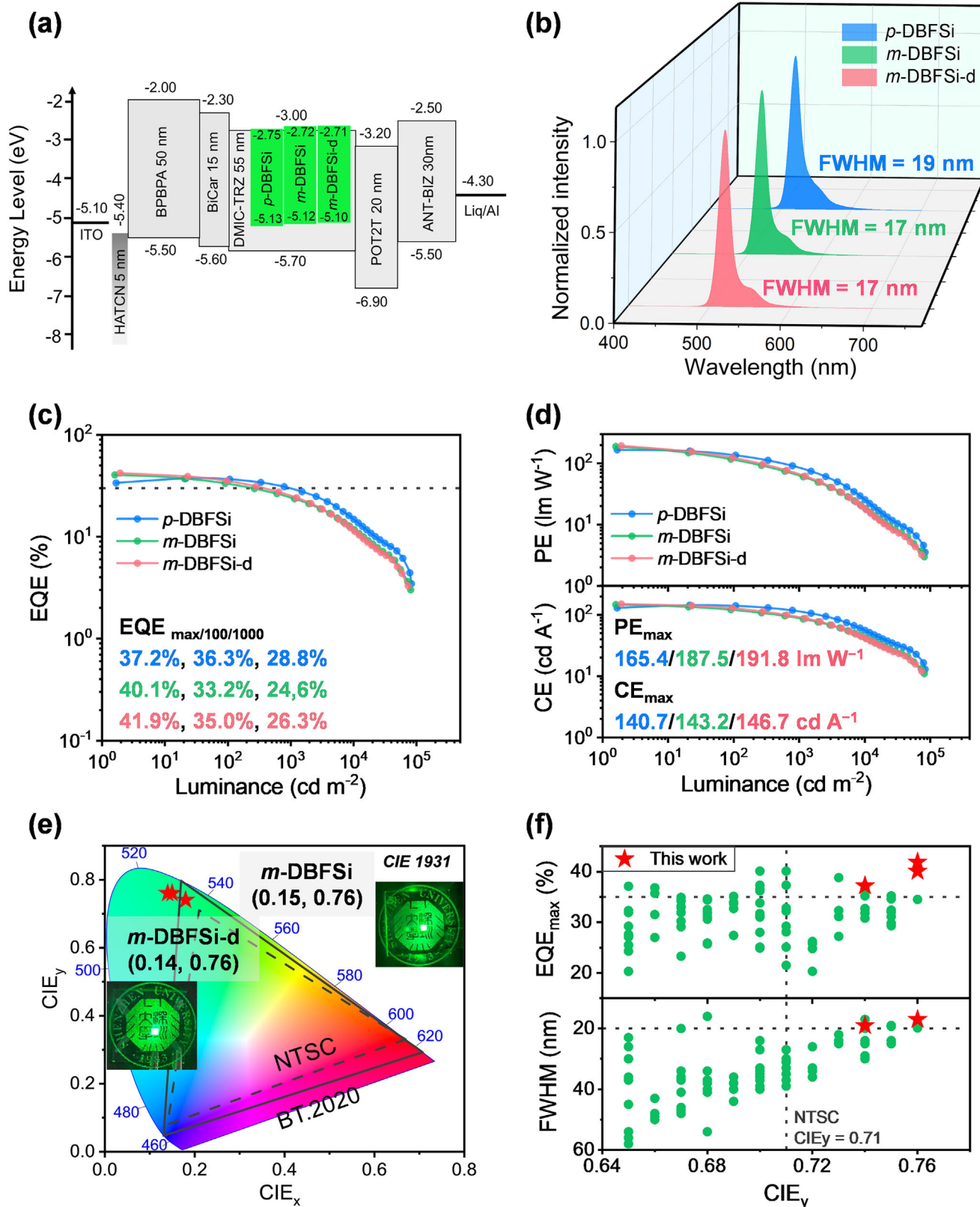


FIGURE 4 | (a) Device architecture and energy level diagram of the functional materials; (b) The EL spectra of the devices (recorded at 1000 cd m⁻²). (c) The EQE-L curve. (d) The PE-L and CE-L curve. (e) CIE coordinates of at 1000 cd m⁻², inset: photograph of *m*-DBFSi-based and *m*-DBFSi-d-based devices. (f) A comparison of EQE_{max} and FWHM versus CIE_y results for the reported green OLEDs based on MR-TADF.

TABLE 2 | Summary of the EL data.

Emitter	V_{on}^{a} [V]	$\lambda_{\text{EL}}^{\text{b}}$ [nm]	FWHM ^b [nm/eV]	$L_{\text{max}}^{\text{c}}$ [cd m ⁻²]	$\text{CE}_{\text{max}}^{\text{d}}$ [cd A ⁻¹]	$\text{PE}_{\text{max}}^{\text{e}}$ [lm W ⁻¹]	EQE^{f} [%]	CIE ^b (x, y)
<i>p</i> -DBFSi	2.4	520	19/0.085	85796	140.7	165.4	37.2/36.3/28.8	(0.18, 0.74)
<i>p</i> -DBFSi- HF	2.2	521	20/0.086	176390	127.2	156.1	34.2/34.1/33.6	(0.19, 0.74)
<i>m</i> -DBFSi	2.4	517	17/0.080	83364	143.2	187.5	40.1/33.2/24.6	(0.15, 0.76)
<i>m</i> -DBFSi- HF	2.2	518	18/0.081	160894	134.3	175.9	37.7/36.8/33.7	(0.18, 0.75)
<i>m</i> -DBFSi-d	2.4	517	17/0.078	76174	146.7	191.8	41.9/35.0/26.3	(0.14, 0.76)
<i>m</i> -DBFSi-d- HF	2.2	518	18/0.080	165592	137.6	184.9	38.6/38.3/36.0	(0.17, 0.75)

^{a)} Turn-on voltage recorded at the luminance of 1 cd m⁻²;

^{b)} Recorded at 1000 cd m⁻²;

^{c)} Maximum brightness;

^{d)} Maximum current efficiency;

^{e)} Maximum power efficiency;

^{f)} EQE at maximum, 100 cd m⁻² and 1000 cd m⁻².

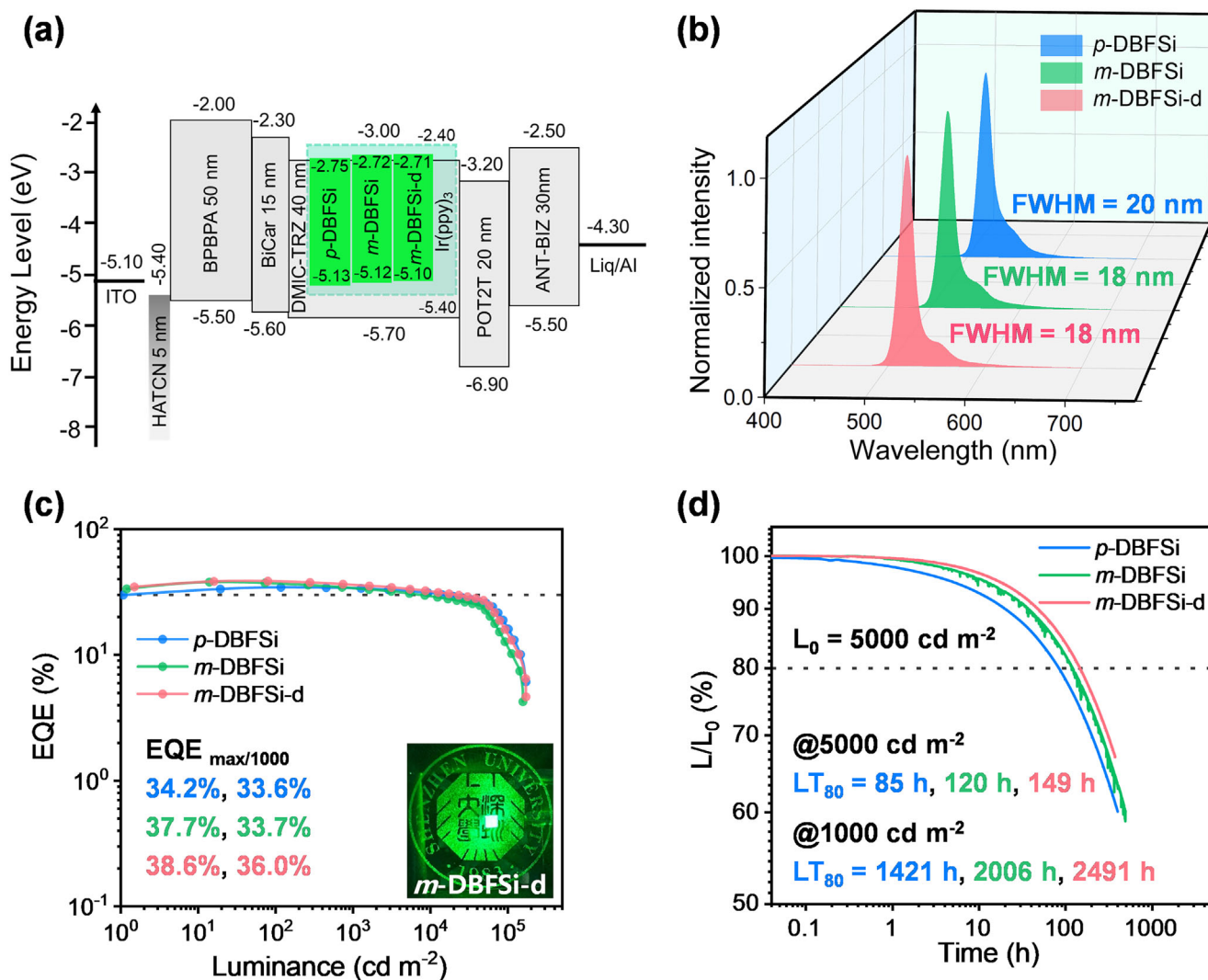


FIGURE 5 | (a) Device architecture and energy level diagram of the functional materials; (b) The EL spectra of the devices (recorded at 1000 cd m⁻²). (c) The EQE-L curve, inset: photograph of *m*-DBFSi-d-based device. (d) Operation lifetime of the sensitized devices at the initial luminance of 5000 cd m⁻².

trend is rationalized by the increased BDEs of the $C_{PAH}-C_{TPS}$ and $C_{PAH}-C_{Si}$ bonds in *meta*-linked emitters compared with *p*-DBFSi (Table S8), which endows the skeleton with greater intrinsic resilience [64, 65]. In addition, benchmarking representative MR-core BDEs against pristine DBFBN indicates that TPS substitution maintains the intrinsic stability of the BN-fused backbone while selectively strengthening key bonds (Table S9), providing a molecular-level rationale for the improved operational durability. Building on this, the partially deuteration in *m*-DBFSi-d provides an additional stabilizing contribution via an isotope effect that reduces high-frequency vibrational activation, consistent with its markedly prolonged device stability.

To further elevate EL performances and fortify device durability, phosphor-sensitized OLEDs were constructed by incorporating an efficient sensitizer. Ir(ppy)₃ was employed owing to its pronounced spectral overlap with these emitters, facilitating efficient Förster resonance energy transfer (Figure S44). The EMLs were subsequently replaced by a ternary doped film with optimized emitters: Ir(ppy)₃: DMIC-TRZ ratio of 2: 10: 88. As illustrated in Figure 5 and Figure S45, the ternary OLEDs deliver narrow pure-green emission with EL peak/FWHM values of 521 nm/20 nm (0.086 eV), 518 nm/18 nm (0.081 eV), and 518 nm/18 nm (0.080 eV) for *p*-DBFSi, *m*-DBFSi and *m*-DBFSi-d, respectively. Although the spectra exhibit slight broadening compared with their binary counterparts, leading to a minor erosion of color purity, the *m*-DBFSi- and *m*-DBFSi-d-based devices retain high CIE y-coordinates of 0.75, squarely within the regime of state-of-the-art pure-green MR-TADF emitters (Table S7). Benefiting from efficient Förster resonance energy transfer and accelerated exciton dynamics, the devices exhibit superior EL performance with greatly suppressed efficiency roll-off. The EQE_{max}, PE_{max}, and CE_{max} reach 34.2%/37.7%/38.6%, 156.1/175.9/184.9 lm W⁻¹, and 127.2/134.3/137.6 cd A⁻¹ for *p*-DBFSi, *m*-DBFSi, and *m*-DBFSi-d, respectively. The modest decrease in EQE_{max} upon sensitization may contribute to the reduced $\Theta_{//}$ of the ternary films (Figure S31), which causes a slightly lowered light-outcoupling efficiency [60, 61]. Remarkably, the operational stability is substantially improved, achieving LT₈₀ lifetimes of 1421 h (*p*-DBFSi), 2006 h (*m*-DBFSi), and 2491 h (*m*-DBFSi-d) at an initial luminance of 1000 cd m⁻². This pronounced improvement is ascribed to the intrinsic chemical robustness of the Ir(ppy)₃ sensitizer coupled with the tetrahedral configuration of the TPS framework, which effectively suppresses undesired Dexter energy transfer (DET) pathways. These lifetimes place the devices among the best-performing ultra-pure green OLEDs based on *meta*-diboron MR emitters and indicate that TPS substitution already imparts substantial devices durability, while additional deuteration of the TPS units affords a further gain in stability, thereby validating this molecular design as an effective way to simultaneously achieve ultra-high color purity, high efficiency, and long operational lifetime.

3 | Conclusions

In summary, this work establishes a robust molecular design strategy that couples π -extension of a *meta*-diboron MR framework with TPS decoration and subsequent deuteration to realize pure-green MR-TADF emitters with superior color purity, exceptional efficiency, and long-term operational stability. Incorporation of a dibenzofuran unit into the *meta*-diboron framework

precisely steers emission into the green region, while peripheral TPS groups further increase color purity, and the tetrahedral configuration of TPS curtails intermolecular coupling and improves stability. The deuteration of TPS units further reduces the zero-point energy and strengthens C—D bonds, leading to better device lifetimes and efficiencies. As a result, OLEDs employing *m*-DBFSi and its deuterated analogue *m*-DBFSi-d achieve exceptional color purity (CIE_y = 0.76), extraordinary EL performance, with EQE_{max} values of 40.1% and 41.9%, and PE_{max} approaching 200 lm W⁻¹, which places them among the best-performing narrowband pure-green OLEDs targeting the BT.2020 standard. Impressively, the *m*-DBFSi-d-based device exhibits remarkably enhanced operational stability, affording an LT₈₀ of 2491 h at an initial luminance of 1000 cd m⁻² and outperforming previous green emitters with similar *meta*-diboron frameworks. Altogether, these findings establish a versatile molecular design paradigm in which TPS substituent and deuteration are used synergistically to address the intertwined challenges of color purity, efficiency, and operational stability in MR-TADF materials for next-generation ultra-high-definition OLED displays.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adma72684-sup-0001-SuppMat.docx.